

Gender Classification Using Neural Networks in R

This project demonstrates how to classify gender based on facial features using neural networks in the R programming language. The dataset used for this project was downloaded from Kaggle and contains the following 7 features:

Dataset Description:

- **long_hair** – Binary column: 1 = long hair, 0 = not long hair
- **forehead_width_cm** – Forehead width in centimeters
- **forehead_height_cm** – Forehead height in centimeters
- **nose_wide** – Binary: 1 = wide nose, 0 = not wide
- **nose_long** – Binary: 1 = long nose, 0 = not long
- **lips_thin** – Binary: 1 = thin lips, 0 = thick lips
- **distance_nose_to_lip_long** – Binary: 1 = large distance between nose and lips, 0 = short distance
- **gender** – Target variable: male or female

Before starting the analysis, the following R packages were installed and loaded:

- **neuralnet** – For building and training neural networks
- **keras & tensorflow** – For deep learning (optional, not used directly in this script)
- **readxl** – For reading Excel files
- **dplyr** – For data manipulation

Data Preprocessing

1. **Loading the data:**

The dataset was imported using the `read_excel()` function.

2. **Basic exploration:**

Summary statistics were generated to understand the distribution of the features.

3. **Data transformation:**

The target column `gender` was converted to a factor, which is necessary for classification tasks in R.

Data Splitting

The dataset was split into a training set (80%) and a test set (20%) using a fixed random seed (`set.seed(113)`) to ensure reproducibility.

Neural Network Training

A neural network was trained using the `neuralnet()` function with the following configuration:

- **Input features:** All columns except *gender*
- **Target:** *gender*
- **Architecture:** Two hidden layers with 4 and 2 neurons respectively
- **Activation function:** Sigmoid (default)
- **Output:** Non-linear (classification problem)

A visualization of the trained network was also created to inspect its structure.

Model Evaluation

The model was evaluated in the following way:

1. Prediction

Predictions were made using the predict() function on the test set.

2. Label transformation

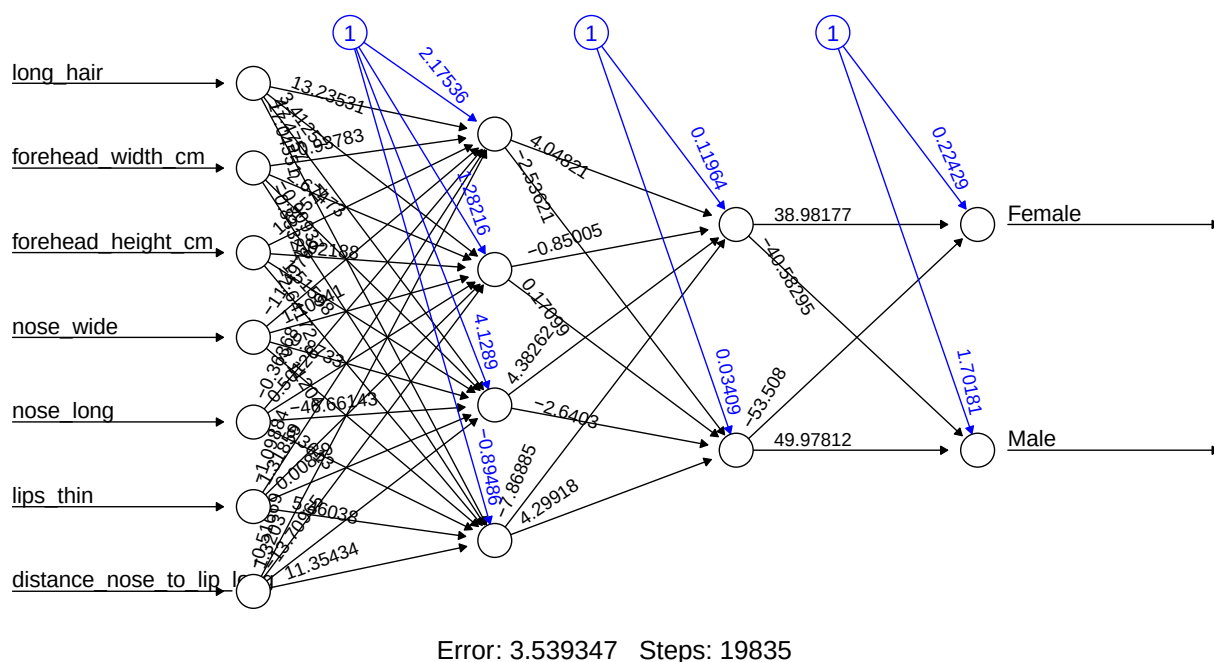
Numeric outputs from the model were mapped to their respective category labels: "Male" and "Female".

3. Confusion matrix

A table was generated comparing the predicted labels with the actual values.

4. Accuracy calculation

The accuracy was calculated by comparing the predicted labels to the actual ones. The model achieved an accuracy of 97.5%.



Final Results

- Predicted (test set):
 - 19 female
 - 21 male
- Actual (test set):
 - 20 female
 - 20 male

The model showed excellent performance, with only one misclassified instance in the test set.

Conclusion

This project shows that even a relatively simple neural network implemented in R can perform very well on a binary classification task using structured data. With a prediction accuracy of 97.5%, the model proves to be highly reliable for this type of classification problem.